

Test Time Adaptation Methods for Point Cloud Registration in Laparoscopic Surgery

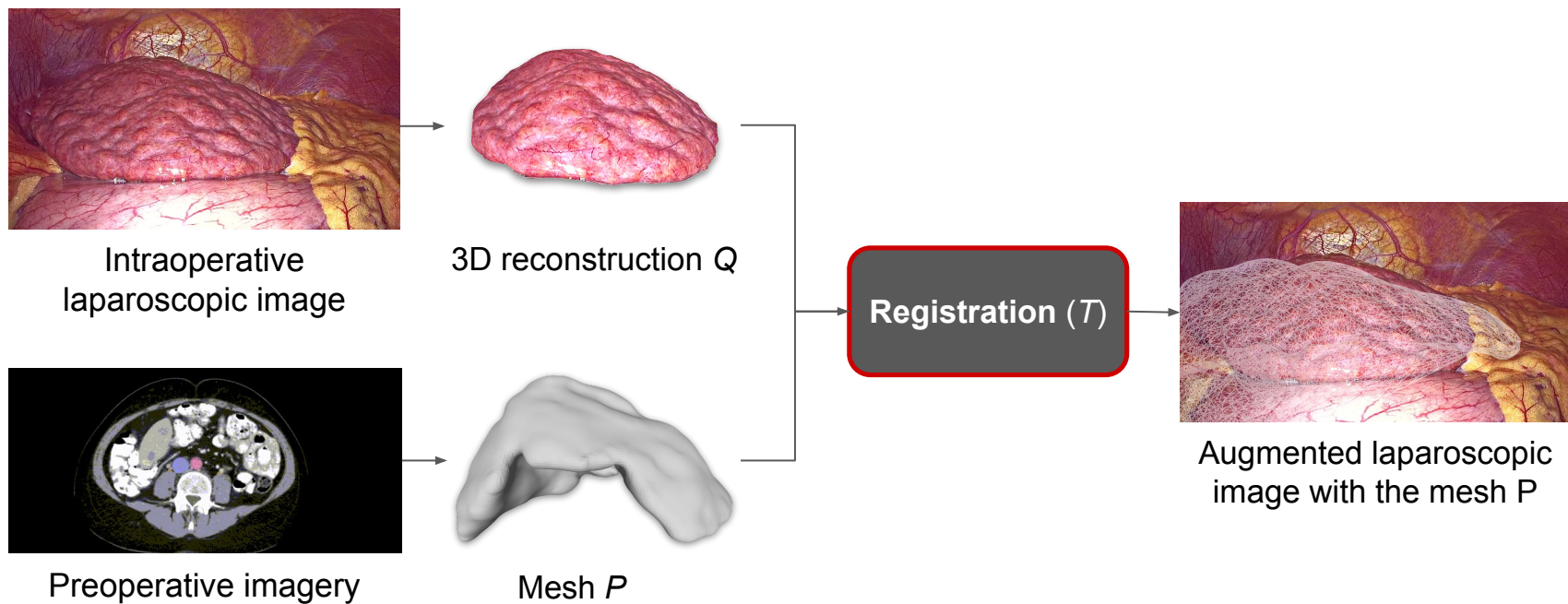
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COLlaborative Intelligence and AAutonomy in
Image-guided S Surgery (COLAS)

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What is registration ?



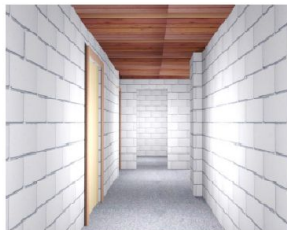
Why registration struggles at test time ?

Train :

Test :

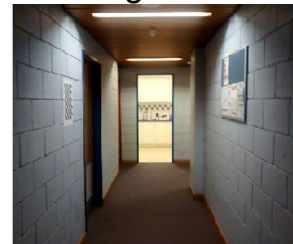
Camera localisation
(Acharya et al., 2019)

Source data, label



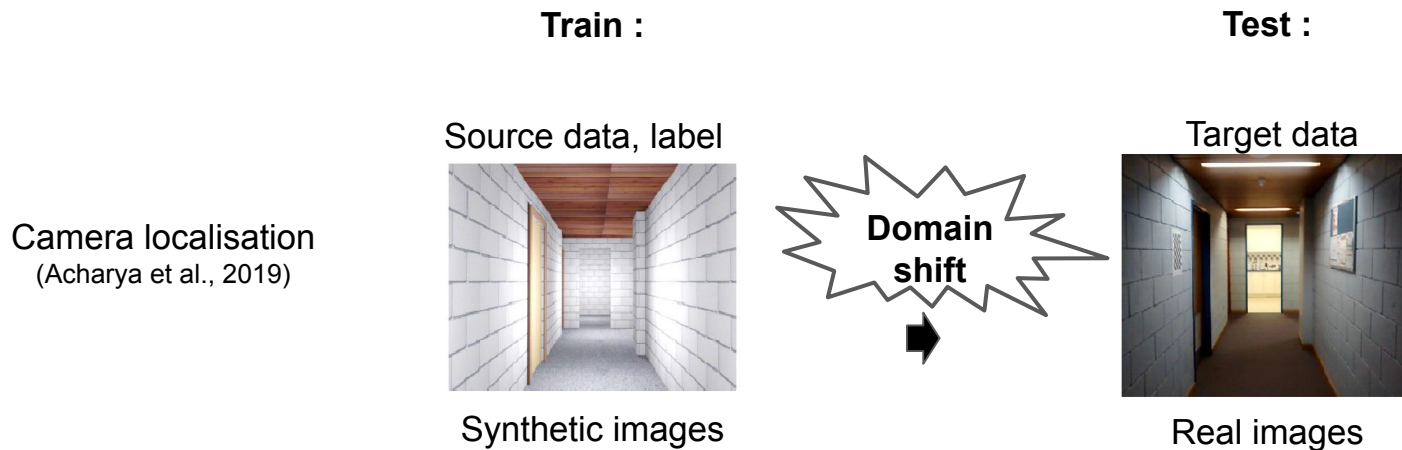
Synthetic images

Target data

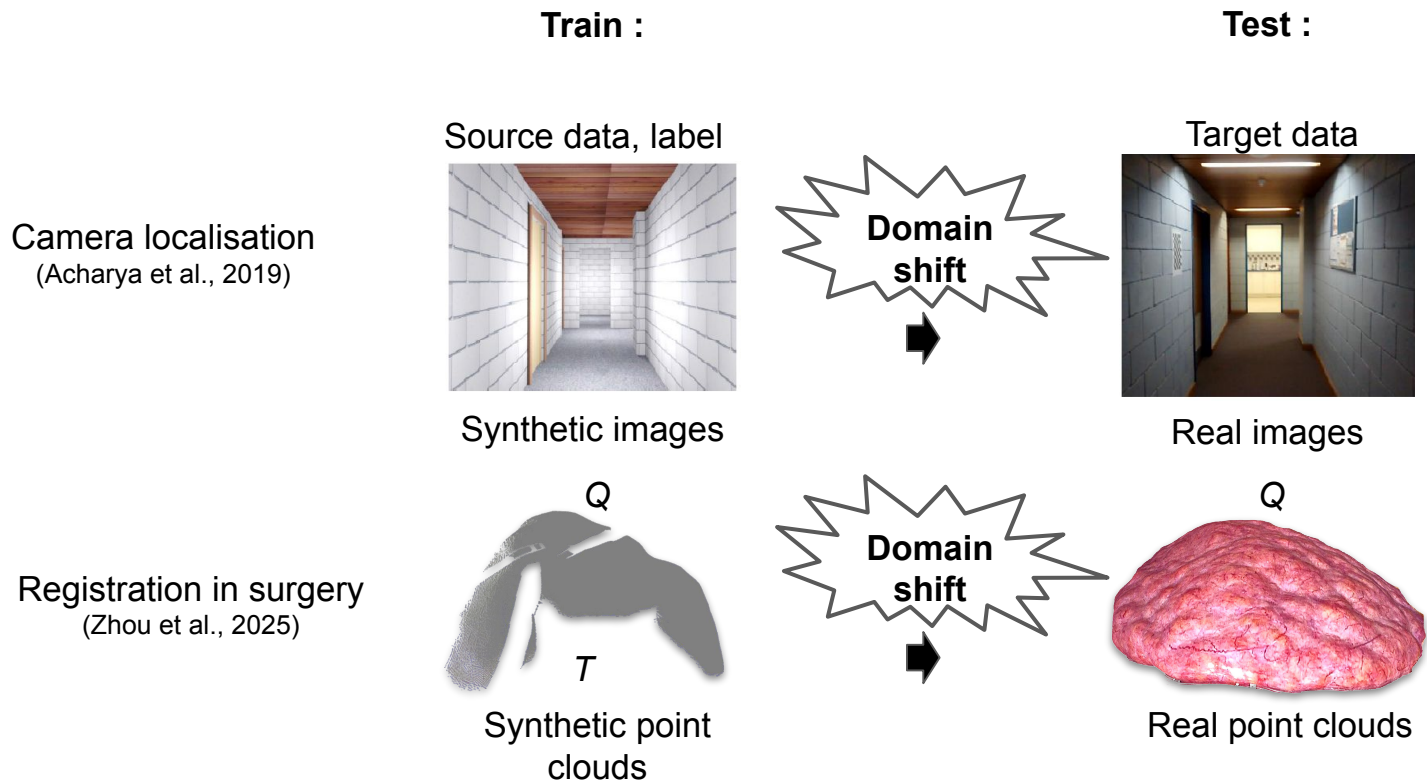


Real images

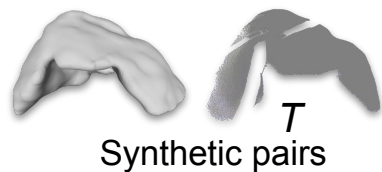
Why registration struggles at test time ?



Why registration struggles at test time ?



→ Test time adaptation (TTA)

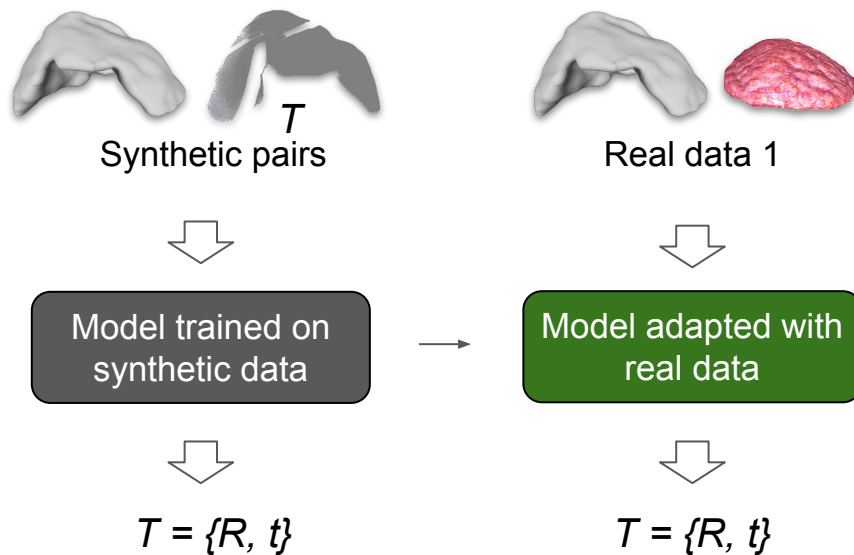


Model trained on
synthetic data

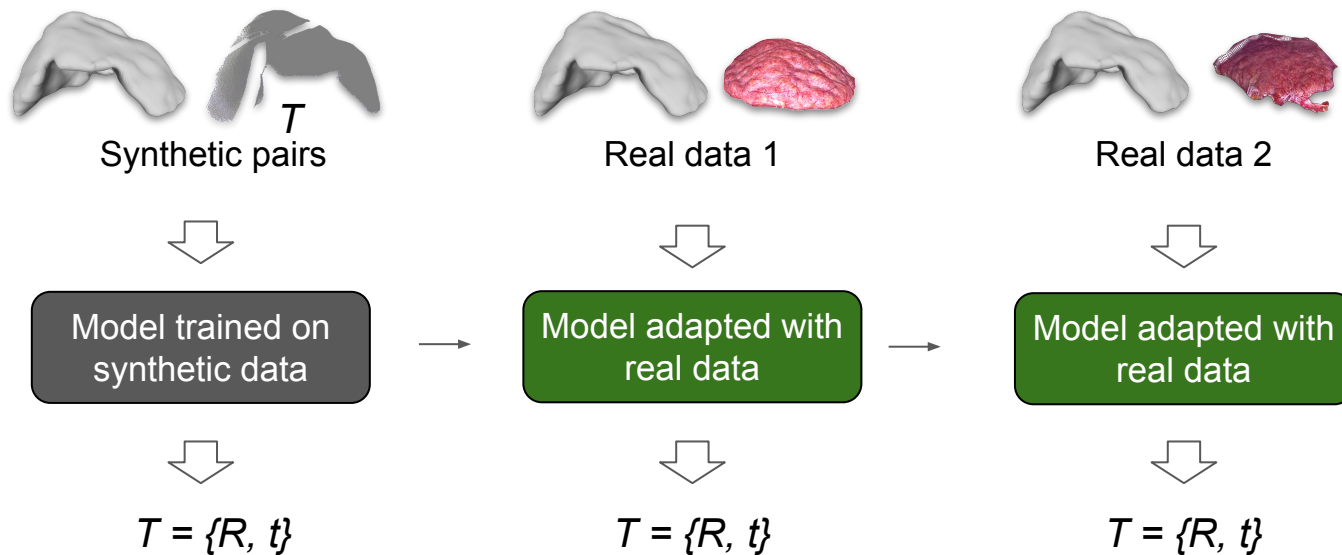


$$T = \{R, t\}$$

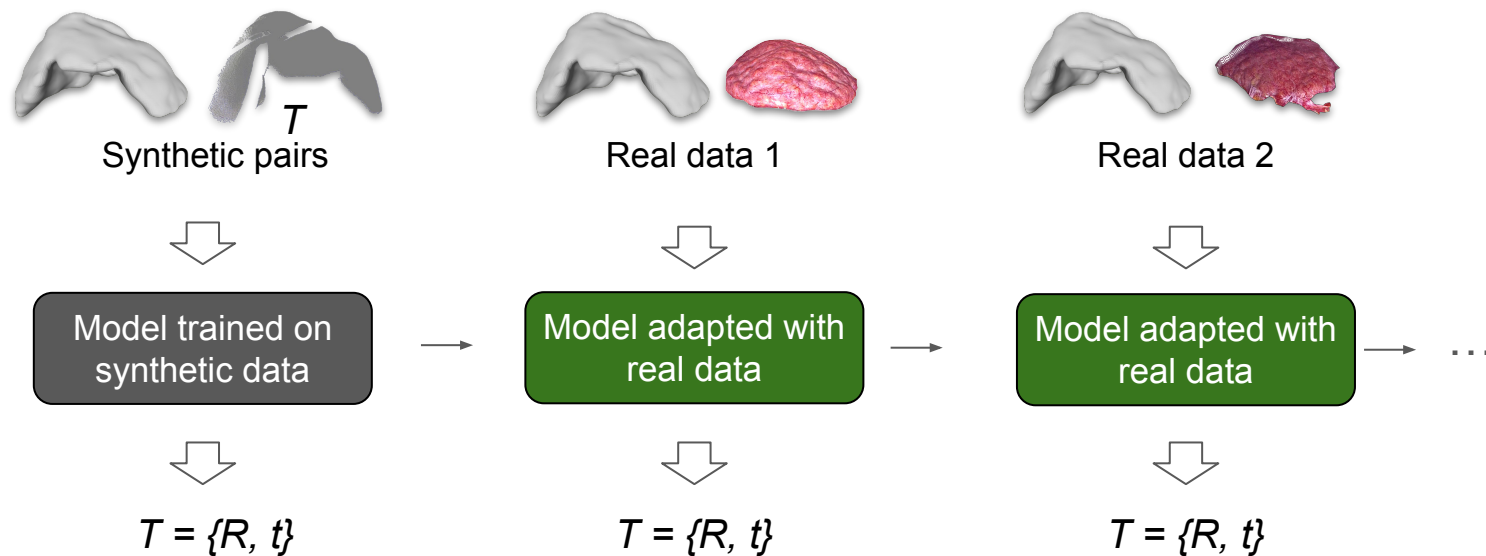
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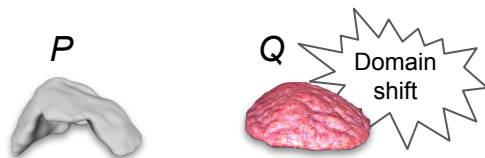


→ Test time adaptation (TTA)



→ TTA for registration in literature :

✗ For one input only



✗ Uses specific mechanisms of classification or segmentation

Contributions :

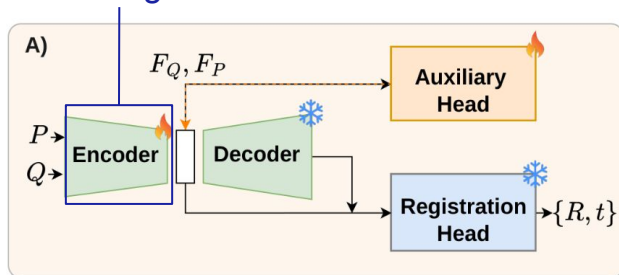
✓ *Compared and modified 3 TTA families for registration:*

A) Model, B) Normalisation and C) Input adaptation

A) Model adaptation

- Point-TTA (Hatem et al., 2026):

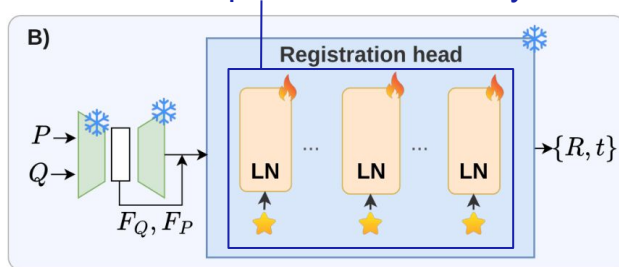
Adapts the weights of the encoder



B) Normalisation adaptation

- LN (Yazdanpanah et al., 2026):

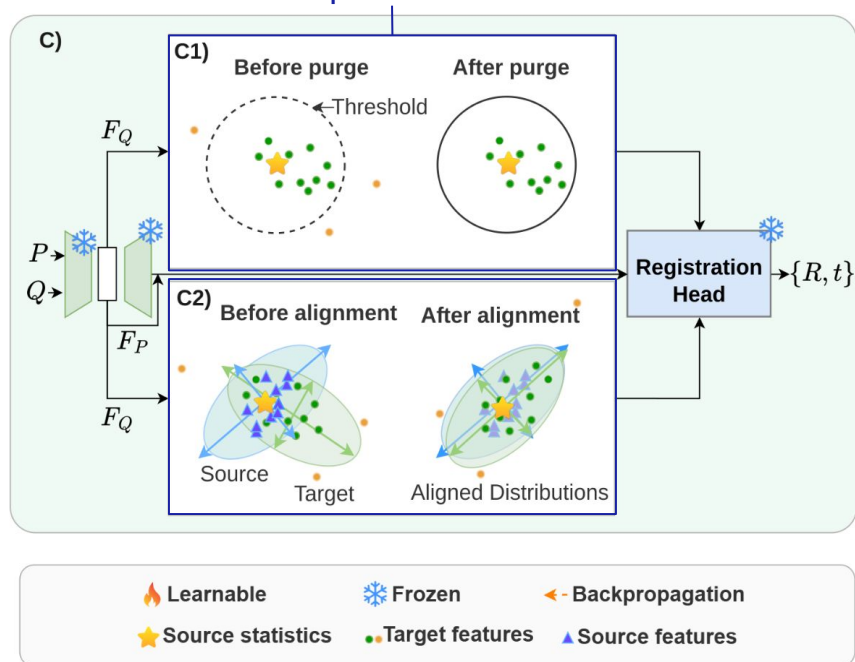
Adapt normalisation layers



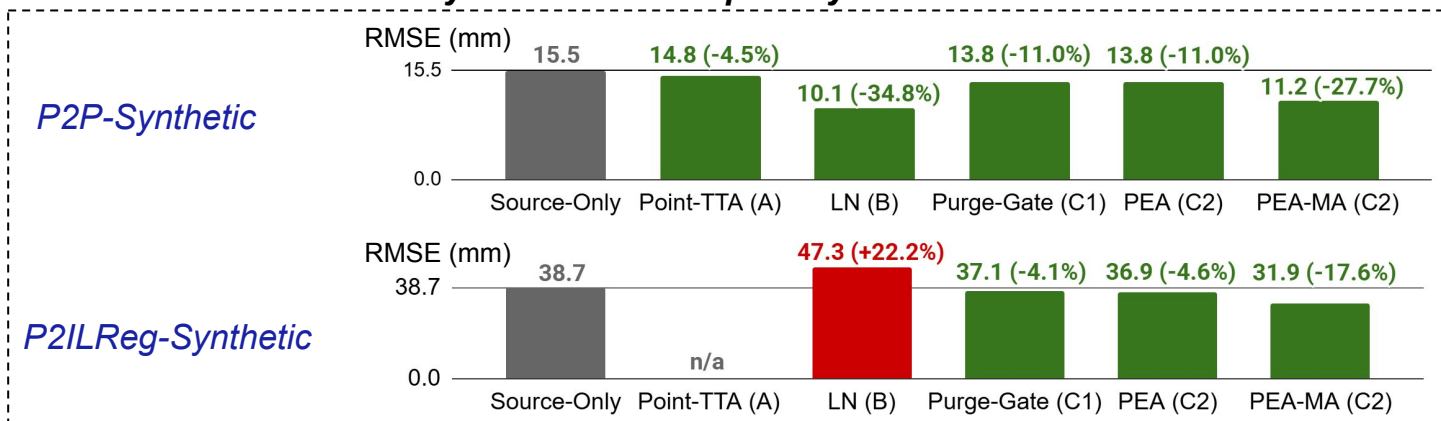
C) Input adaptation:

- C1) Purge-Gate (Yazdanpanah et al., 2026),
- C2) PEA (Ma, et al., 2026), PEA-MA

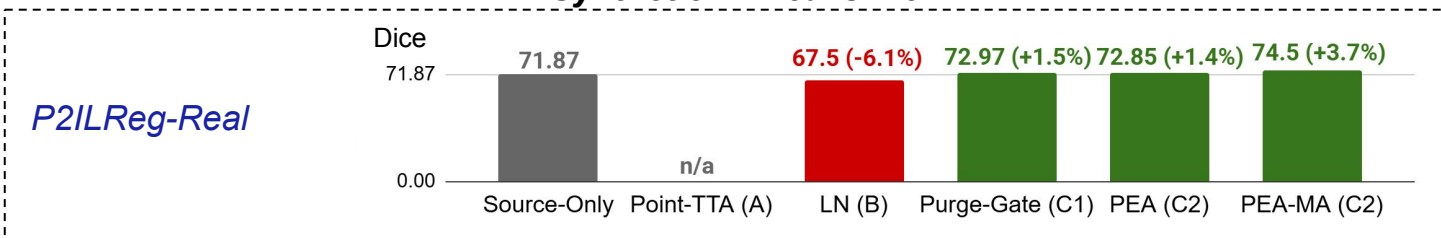
Adapt features



Synthetic → Corrupted synthetic shift



Synthetic → Real shift



Source-Only: PARE-Net (Yao et al., 2024)

Time per frame : 162 → 1226 ms (A); 184-414 ms (B,C)

✓ **Test time adaptation can improve laparoscopic registration**

- Input adaptation : robust and fast
- Normalisation adaptation : unstable across datasets
- Model adaptation : too slow for the operating room

Paper



Code



Contact

